Mechanics WS 5

(Take $g = 9.8\ m \cdot s^{-2}$, for the following Q1-4)

1. An object of mass m is thrown vertically upwards in a medium with a resistance $R = \frac{mv}{k}$, where v is the velocity of the object and k is a constant. Given $t = 0 = x$ and the initial velocity is $k(c - g)$, where c is a constant and g is the acceleration due to gravity, x is the distance travelled in time t.
   a. Draw a neat sketch of the motion and the forces acting at a point P, distance x from the origin. Hence write down the equation of the motion.
   b. Find the time taken by the object to reach the highest point H and find the height of H above the point of projection.
   c. The object falls to its original position with the same law of resistance. Will the time of descent be the same as that of scent? Give your reasons.
2. Find the tangential speed of the earth at the equator due to its rotation about its axis, given that the radius of the equatorial circle is $6440\ km$. [Use exercise 4(a)]
3. A ball of mass $3kg$ is thrown vertically upwards from the origin with an initial speed of $6\ ms^{-1}$. The ball is subject to a downward gravitational force of 18 newtons and air resistance of $\frac{v^2}{2}$ newtons in the opposite direction to the velocity, v metres per second
   a. Show that until the ball reaches its highest point, the equation of motion is
   $\ddot{y} = -\frac{1}{6}(v^2 + 36)$ where y metres is its height
   b. Show that while the ball is rising $v^2 = 36\left(2e^{-\frac{y}{3}} - 1\right)$
   c. Show that the ball's maximum height is $3 \ln 2$ metres
   d. Show that the ball's speed on its return to the origin is $3\sqrt{2}$ metres per second
4. Two stones are thrown simultaneously from the same point in the same direction and with the same non-zero angle of projection (upward inclination to the horizontal), α, but with different velocities V_1 and V_2 metres per second where $V_1 < V_2$.
   The slower stone hits the ground at a point P on the same level as the point of projection. At that instant the faster stone just clears a wall of height h metres above the level of projection and its (downward) path makes an angle β with the horizontal.
   a. Show that, while both stones are in flight, the line joining them has an inclination to the horizontal which is independent of time
   b. Hence, express the horizontal distance from P to the foot of the wall in terms of h and α.
   c. Show that $V_2\ (\tan\alpha + \tan\beta) = 2V_1 \tan\alpha$ and deduce that, if $\beta = \frac{1}{2}\alpha$, then
   $$V_1 < \frac{3}{4}V_2$$
5. A particle moves along a straight line so that its displacement, x metres, from a fixed point O is given by $x = 1 - 3\cos(\frac{t}{2})$, where t is measured in seconds.
   a. Sketch the graph of x as a function of t for $0 \le t \le 4\pi$
   b. Hence, or otherwise, find when and where the particle first comes to rest after $t = 0$.
   c. Find a time when the particle reaches its maximum speed. What is this speed?
6. In a certain harbour at 6am it is low tide and at 12:30pm it is high tide. Low tide is at 4.5 metres and high tide is 10.6 metres. Assume that the motion of the tide can be approximated by SHM.
   a. Write down an equation to represent the depth of water at time, t.
   b. If a ship requires a minimum depth of 8 metres, when can it first enter the harbour after 6am?
7. The depth x metres of the water in a certain South Coast harbour is found to vary approximately according to the equation $\ddot{x} = -\frac{x}{4}$. Given that t is the time in hours and it is known that the difference between high and low tide is 4 metres,

a. Prove that the time between successive high tides is 12.6 hours, correct to the nearest $\frac{1}{10}$ of an hour.

b. Find the rise in the water level during the first hour after low tide. Give your answer in metres, correct to two decimal places.

c. Find the rate at which the level is falling two hours after high tide. Give your answer in metres per hour, correct to two decimal places.

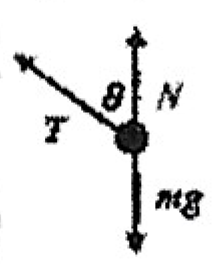
8. The velocity v m/s of a particle moving in simple harmonic motion along the x-axis is given by $v^2 = -5 + 6x - x^2$, where x is in metres.
   a. Between which two points is the particle oscillating?
   b. Find the centre of motion of the particle.
   c. Find the maximum speed of the particle.
   d. Find the acceleration of the particle in terms of x.

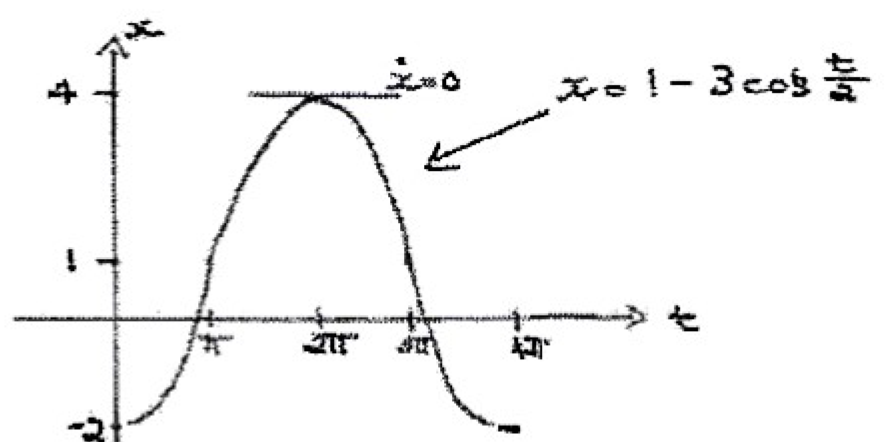
9. A particle P moves along the x-axis so that at time t seconds it is $x$cm from the origin O and its velocity is v cm/s. Initially the particle is at rest at the origin.
   a. If the acceleration of P is given by $\ddot{x} = 4(40 - x)$ cm²/s, use $\ddot{x} = \frac{d}{dx}(\frac{1}{2}v^2)$ to show that $v^2 = 4(80x - x^2)$
   b. Prove that P moves in the interval $0 \leq x \leq 80$.
   c. Find the maximum velocity of the particle, and where the maximum occurs.

10. A particle is moving in simple harmonic motion about the origin
   a. Assuming that $\ddot{x} = -n^2 x$, show that $v^2 = n^2(a^2 - x^2)$, where a is the amplitude
   b. When the particle is 3 metres from the origin, its speed is 8m/s, and when it is 4 metres from the origin its speed is 6 m/s. Find the period and amplitude of the motion.
   c. Find the greatest acceleration of the particle

11. A particle is moving in a straight line. At time t seconds it has displacement x metres from a fixed point O on the line. Its velocity v m/s is given by $v = -\frac{1}{8}x^3$. The particle is initially 2 metres to the right of O.
   a. Show that the acceleration a, is given by: $a = \frac{3}{64}x^5$
   b. Find an expression for x in terms of t
   c. Describe the limiting motion of the particle

Answers:

1. (a) $m\frac{dv}{dt} = -mg - \frac{mv}{k}$ (b) $t = k\log_e(c/g), h = k^2(c-g) + gk^2\log_e(g/c)$ (c) The time of descent < the time of ascent, since the gravity opposes the upward motion of the object.
2. $468 m/s$ $(1690\ km/h)$
3. (a) 

(e) the particle would lift off the table and behave like a regular conical pendulum
4. .
5. a.


b. $\dot{x} = 0, t = 2\pi$ s when $x = 4$m; c. $t = \pi$ s, max speed= $\frac{3}{2}$ m/s
6. a. $x = 7.55 - 3.05\cos(\frac{2\pi}{13}t)$; b. 9:33 am
7. b. 0.24m; c. 0.84 m/h
8. a. $1 \leq x \leq 5$; b. 3 m; c. 2 m/s; d. $3 - x$ m/s²
9. c. 80 m/s when $x = 40$
10. b. $A = 5,\ T = \pi^2$; c. $max\ acceleration = 20\ ms^{-2}$
11. b. $x = \frac{2}{\sqrt{t+1}}$; c. $As\ t \to 0, x \to 0, v \to 0\ ms^{-1}$